Model Evaluation

ADS 503 Final Team Project - Diabetes Health Indicators (BRFSS 2015)

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Libraries

```
library(tidyverse)
library(caret)
library(recipes)
library(pROC)
library(scales)
library(doParallel)
library(glmnet)
library(ggplot2)
library(ranger)
library(gbm)
library(PRROC) # Explicitly loaded for Precision-Recall evaluation metrics

set.seed(503)
dm_palette <- c("No Diabetes" = "#4C9F70", "Diabetes" = "#D1495B")
```

Importing from Model Tuning

```
setwd(r"(C:\Users\User\ADS503_Team5\Raw Data)")


load("tuned_models_and_data.RData")
#source: https://www.rdocumentation.org/packages/base/versions/3.6.2/topics/load
```

ROC Predicted Model Construction

```
#Source: Robin (2026). Package 'pROC'.
#Link:  https://xrobin.github.io/pROC/
prob_lda <- predict(mod_glm, x_test_processed, type = "prob")[["Diabetes"]] #gemini suggeste
prob_enet <- predict(mod_glmnet, x_test_processed, type = "prob")[["Diabetes"]] #gemini sugg
prob_rf    <- predict(mod_rf, x_test_df, type = "prob")[["Diabetes"]]
#gemini suggested [["Diabetes"]] subset when calling predict()
prob_gbm    <- predict(mod_gbm, x_test_df, type = "prob")[["Diabetes"]]
#gemini suggested [["Diabetes"]] subset when calling predict()
```

Model ROC construction

```
roc_lda  <- roc(test_y, prob_lda, levels = c("NoDiabetes", "Diabetes"), quiet = TRUE)
roc_enet    <- roc(test_y, prob_enet, levels = c("NoDiabetes", "Diabetes"), quiet = TRUE)
roc_rf  <- roc(test_y, prob_rf, levels = c("NoDiabetes", "Diabetes"), quiet = TRUE)
roc_gbm     <- roc(test_y, prob_gbm, levels = c("NoDiabetes", "Diabetes"), quiet = TRUE)
```

Delong Test Results: baseline comparison

```
# Execute statistical comparison of holdout curves
#LDA Vs. Elastic Net
delong_result <- roc.test(roc_lda, roc_enet, method = "delong")
print(delong_result)
```

```

    DeLong's test for two correlated ROC curves

data:  roc_lda and roc_enet
Z = -0.45462, p-value = 0.6494
alternative hypothesis: true difference in AUC is not equal to 0
95 percent confidence interval:
 -0.0001649040  0.0001028077
sample estimates:
AUC of roc1 AUC of roc2
  0.8272879   0.8273189
```

```
#LDA vs. GBM
delong_result2 <- roc.test(roc_lda, roc_gbm,method = "delong")
print(delong_result2)
```

```
    DeLong's test for two correlated ROC curves

data:  roc_lda and roc_gbm
Z = -3.0315, p-value = 0.002433
alternative hypothesis: true difference in AUC is not equal to 0
95 percent confidence interval:
 -0.004740708 -0.001017740
sample estimates:
AUC of roc1 AUC of roc2
  0.8272879   0.8301671
```

```
#LDA versus Random Forest
delong_result3 <- roc.test(roc_lda, roc_rf,method = "delong")
print(delong_result3)
```

```
    DeLong's test for two correlated ROC curves

data:  roc_lda and roc_rf
Z = -1.1642, p-value = 0.2443
alternative hypothesis: true difference in AUC is not equal to 0
95 percent confidence interval:
 -0.003791111  0.000965619
sample estimates:
AUC of roc1 AUC of roc2
  0.8272879   0.8287006
```

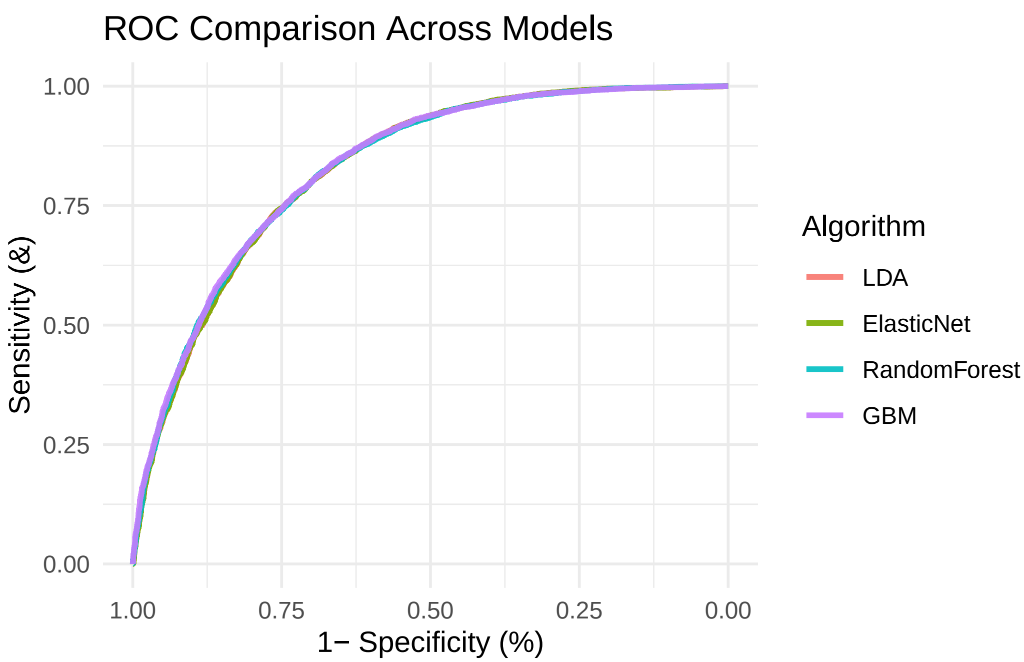
Judging by the results of the Delong's tests, most of the models remain statistically indistinguishable from each other except for LDA and GBM, where the p value of that test is 0.002433. GBM outperforms LDA significantly.

ROC Comparison

```
##Source: Robin (2026). Package 'pROC'.
#Link:  https://xrobin.github.io/pROC/
# Build individual ROC objects first
r_lda   <- roc(test_y, prob_lda,   levels = c("NoDiabetes", "Diabetes"), quiet = TRUE)
r_enet <- roc(test_y, prob_enet, levels = c("NoDiabetes", "Diabetes"), quiet = TRUE)
r_rf    <- roc(test_y, prob_rf,    levels = c("NoDiabetes", "Diabetes"), quiet = TRUE)
r_gbm   <- roc(test_y, prob_gbm,   levels = c("NoDiabetes", "Diabetes"), quiet = TRUE)
```

```r
# Put them in a named list
#suggested by Gemini:
roc_list <- list(LDA = r_lda, ElasticNet = r_enet, RandomForest = r_rf, GBM = r_gbm)

# Plot all curves simultaneously with ggplot2
#source : https://www.rdocumentation.org/packages/ggROC/versions/1.0/topics/ggroc
ggroc(roc_list, aes = "color", linewidth = 1, alpha = 0.9) +
  theme_minimal() +
  labs(title = "ROC Comparison Across Models", color = "Algorithm", x = "1- Specificity (%)"
```

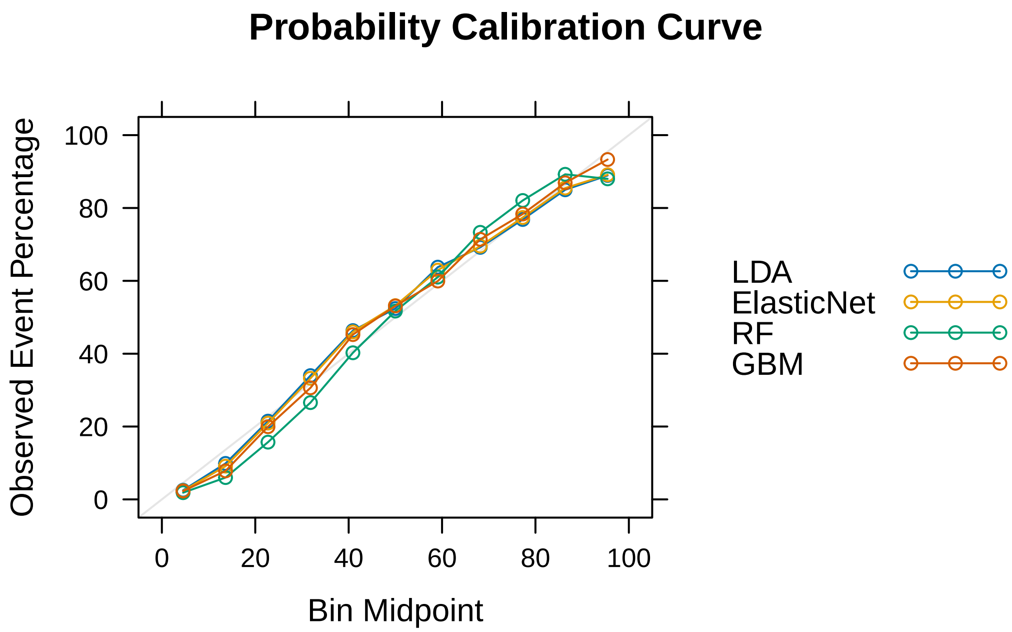


Probability Calibration Assessment

```r
# Create a dedicated dataframe for calibration
calibration_df <- data.frame(
  obs        = test_y,
  LDA        = prob_lda,
  ElasticNet = prob_enet,
  RF         = prob_rf,
  GBM        = prob_gbm
) #suggested by Gemini

#source: Kuhn (2026). Package 'caret': 'calibration'".
#https://cran.r-project.org/web/packages/caret/caret.pdf
```

```r
#source for auto.key
#https://stackoverflow.com/questions/10828016/lattice-auto-key-how-to-adjust-lines-and-points
cal_obj <- calibration(obs ~ LDA+ElasticNet+RF+GBM, data = calibration_df, cuts = 11)
plot(cal_obj, type = "b", main = "Probability Calibration Curve", auto.key = TRUE)
```



Model summaries

```r
#Creating a list of the models loaded in
#source: https://www.w3schools.com/r/r_lists.asp

models <- list(
  LDA      = mod_glm,
  ElasticNet    = mod_glmnet,
  RandomForest = mod_rf,
  GBM          = mod_gbm
)
#Using resamples to give a summary of the models
#https://www.rdocumentation.org/packages/caret/versions/6.0-92/topics/resamples
res <- resamples(models)
summary(res)
```

```
Call:
```

```
summary.resamples(object = res)
```

```
Models: LDA, ElasticNet, RandomForest, GBM
Number of resamples: 5

ROC
                  Min.   1st Qu.    Median      Mean   3rd Qu.      Max. NA's
LDA          0.8196668 0.8212530 0.8237375 0.8240489 0.8259243 0.8296629    0
ElasticNet   0.8197351 0.8213236 0.8237687 0.8240887 0.8258696 0.8297466    0
RandomForest 0.8213277 0.8240323 0.8240529 0.8259629 0.8281790 0.8322225    0
GBM          0.8216358 0.8252026 0.8255894 0.8265866 0.8293572 0.8311482    0

Sens
                  Min.   1st Qu.    Median      Mean   3rd Qu.      Max. NA's
LDA          0.7598585 0.7671499 0.7675035 0.7677618 0.7685234 0.7757737    0
ElasticNet   0.7584439 0.7664427 0.7669731 0.7673728 0.7697613 0.7752431    0
RandomForest 0.7842617 0.7854996 0.7896040 0.7905364 0.7929632 0.8003537    0
GBM          0.7709991 0.7736516 0.7756365 0.7761432 0.7766973 0.7837312    0

Spec
                  Min.   1st Qu.    Median      Mean   3rd Qu.      Max. NA's
LDA          0.7253758 0.7282051 0.7282532 0.7286486 0.7299735 0.7314356    0
ElasticNet   0.7259063 0.7282051 0.7285588 0.7284364 0.7287836 0.7307284    0
RandomForest 0.7024399 0.7082228 0.7122900 0.7101179 0.7137553 0.7138815    0
GBM          0.7161804 0.7197171 0.7238331 0.7230257 0.7256011 0.7297966    0
```

Confusion Matrix for GBM

```r
#Using if-else for the confusion matrix
#source: https://www.rdocumentation.org/packages/base/versions/3.6.2/topics/ifelse

pred_class_gbm <- factor(
  ifelse(prob_gbm > 0.5, "Diabetes", "NoDiabetes"),
  levels = c("NoDiabetes", "Diabetes")
) #Gemini suggested as.factor(), ifelse() and "levels" to make sizes the same for reference a

actual_class <- factor(test_y, levels = c("NoDiabetes", "Diabetes")) #gemini suggested facto

# Evaluate exact metrics
gbm_cm <- confusionMatrix(data = pred_class_gbm, reference = actual_class, positive = "Diabet
print(gbm_cm)
```

```
Confusion Matrix and Statistics
```

```
            Reference
Prediction   NoDiabetes Diabetes
  NoDiabetes       5132     1592
  Diabetes         1937     5477

               Accuracy : 0.7504
                 95% CI : (0.7432, 0.7575)
    No Information Rate : 0.5
    P-Value [Acc > NIR] : < 2.2e-16

                  Kappa : 0.5008

 Mcnemar's Test P-Value : 7.009e-09

            Sensitivity : 0.7748
            Specificity : 0.7260
         Pos Pred Value : 0.7387
         Neg Pred Value : 0.7632
             Prevalence : 0.5000
         Detection Rate : 0.3874
   Detection Prevalence : 0.5244
      Balanced Accuracy : 0.7504

       'Positive' Class : Diabetes
```

Highest Mean ROC: GBM Highest Mean Sensitivity: Random Forest Highest Mean Specificity: LDA

Model Comparison

```r
bestTunes <- list(
  LDA      = mod_glm$bestTune,
  ElasticNet   = mod_glmnet$bestTune,
  RandomForest = mod_rf$bestTune,
  GBM          = mod_gbm$bestTune
)
print(bestTunes)
```

```
$LDA
  parameter
```

```
1      none

$ElasticNet
   alpha      lambda
27   0.4 0.002674312

$RandomForest
  mtry splitrule min.node.size
1    2      gini             1

$GBM
  n.trees interaction.depth shrinkage n.minobsinnode
1     150                 1       0.1             50
```

```
#Using diff to give a summary of the resamples
#https://www.rdocumentation.org/packages/caret/versions/6.0-92/topics/resamples
resamples_diff <- diff(res) #gemini suggested using diff() on the entire res (resample) varia
summary(resamples_diff) #gemini suggested the use of summary() for the final results
```

```
Call:
summary.diff.resamples(object = resamples_diff)

p-value adjustment: bonferroni
Upper diagonal: estimates of the difference
Lower diagonal: p-value for H0: difference = 0

ROC
             LDA     ElasticNet RandomForest GBM
LDA                  -3.983e-05 -1.914e-03   -2.538e-03
ElasticNet   1.00000            -1.874e-03   -2.498e-03
RandomForest 0.07721 0.08125                 -6.238e-04
GBM          0.03812 0.04356    1.00000

Sens
             LDA      ElasticNet RandomForest GBM
LDA                    0.000389  -0.022775    -0.008381
ElasticNet   1.000000            -0.023164    -0.008770
RandomForest 0.001598 0.003055                 0.014393
GBM          0.006523 0.019979   0.001068

Spec
```

```
             LDA     ElasticNet RandomForest GBM
LDA                   0.0002122  0.0185307    0.0056230
ElasticNet   1.00000             0.0183185    0.0054108
RandomForest 0.01421 0.01206                 -0.0129078
GBM          0.20535 0.31973    0.06785
```

After evaluating the summary of differences between models. There are multiple conclusions that may be drawn. ROC:

LDA vs. GBM : p value is 0.03812. Difference is -2.538e-03. This is statistically significant (P<0.05) and despite that small difference in ROC, GBM does outperform LDA.

Elastic Net vs. GBM: p-value is 0.04356. Difference is -2.498e-03. This difference in overall performace is statistically significant (P<0.05) and GBM does outperform elastic net.

Sensitivity: GBM vs. LDA: p-value is 0.006523. Difference is -0.008381. The result is statistically significant (P<0.05) and despite the small difference, GBM does outperform LDA in sensitivity.

Random Forest vs. LDA: p-value is 0.001598 Difference is -0.022775 The result is statistically significant (P<0.05) and despite the small difference, Random Forest does outperform LDA in sensitivity.

Random Forest vs. Elastic Net: p-value is 0.003055 Difference is -0.023164 The result is statistically significant (P<0.05) and despite the small difference, Random Forest does outperform Elastic Net in sensitivity.

GBM vs. Elastic Net: p-value is 0.019979 Difference is -0.008770 The result is statistically significant (P<0.05) and despite the small difference, GBM does outperform Elastic Net in sensitivity.

GBM vs. Random Forest: p-value is 0.001068 Difference is -0.0129078 The result is statistically significant (P<0.05) and despite the small difference, GBM does outperform Random Forest in sensitivity.

Specificity: Random Forest vs. LDA: p-value is 0.01421 . Difference is 0.0185307. The result is statistically significant (P<0.05) and despite the small difference, LDA does outperform Random Forest in specificity

Random Forest vs. Elastic Net: p-value is 0.01206 . Difference is 0.0183185 The result is statistically significant (P<0.05) and despite the small difference, Elastic Net does outperform Random Forest in specificity

Exporting the models to model_evaulation

```r
# Set working directory to ensure it saves to a project data folder
setwd(r"(C:\Users\User\ADS503_Team5\Raw Data)")
# Save evaluation file
save(
  mod_glm, mod_glmnet, mod_rf, mod_gbm, models, gbm_cm,
  bestTunes, calibration_df, test_y, resamples_diff, res, roc_list,
  file = "evaluated_models.RData"
)
```